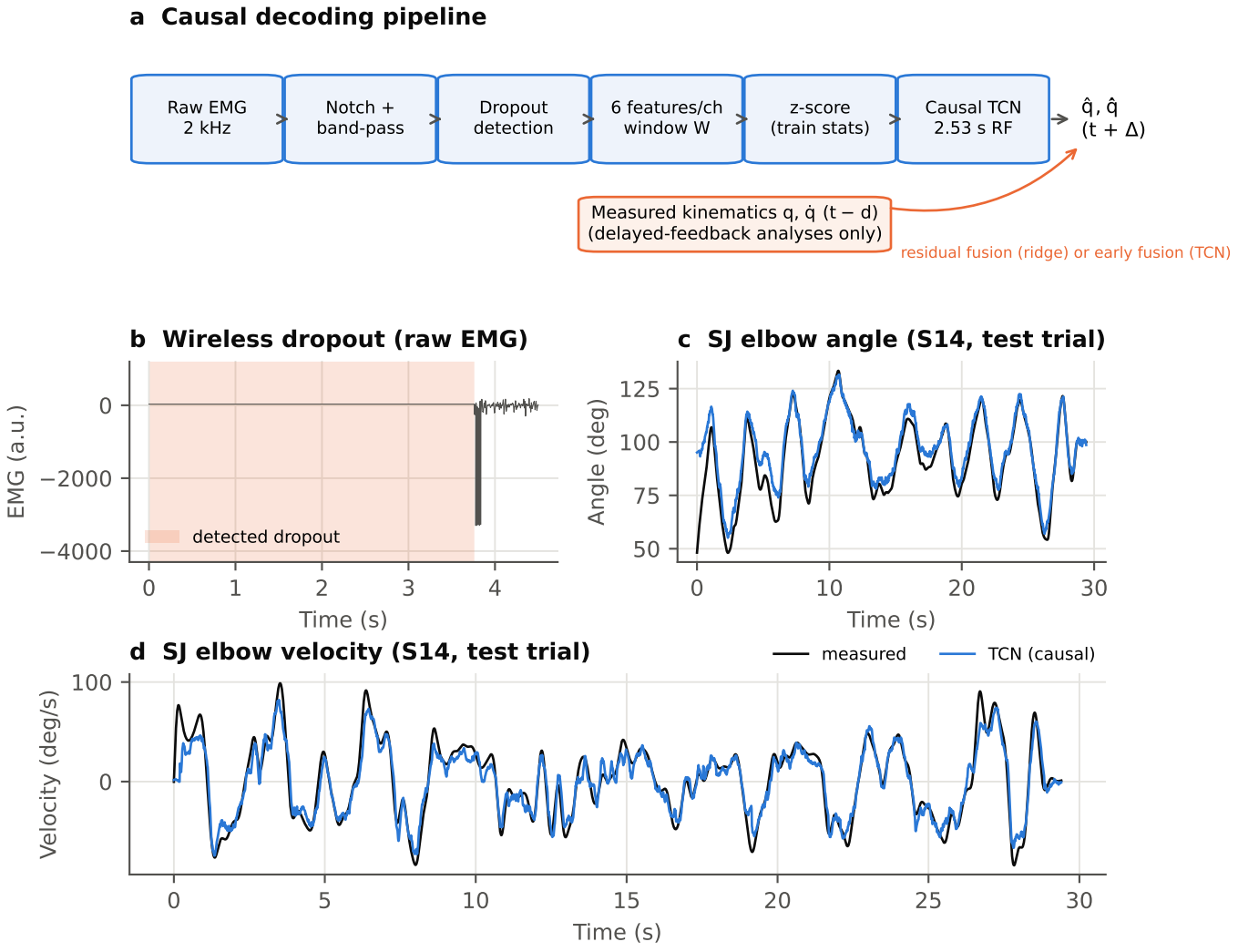

a Causal decoding pipeline
Raw EMG 2 kHz
Notch + band-pass
Dropout detection
6 features/ch window W
z-score (train stats)
Causal TCN 2.53 s RF
q̂, q̂̇ (t + Δ)
Measured kinematics q, q̇ (t − d)
(delayed-feedback analyses only)
residual fusion (ridge) or early fusion (TCN)
b Wireless dropout (raw EMG)
EMG (a.u.)
0
−2000
−4000
detected dropout
0
1
2
3
4
Time (s)
c SJ elbow angle (S14, test trial)
Angle (deg)
125
100
75
50
0
10
20
30
Time (s)
d SJ elbow velocity (S14, test trial)
measured
TCN (causal)
Velocity (deg/s)
100
0
0
5
10
15
20
25
30
Time (s)